Technical ReportProgressive Visual Analytics (PVA) for Incident Management

This document provides a detailed technical description of the Progressive Visual Analytics system developed for forensic analysis of the Incident Management Process dataset.

1. System Architecture

The system uses Progressivis for incremental processing, Plotly for interactive visualization, and IncrementalPCA from scikit-learn for dimensionality reduction. The architecture supports streaming updates and coordinated visual interaction.

2. Dimensionality Reduction (Incremental PCA)

Feature vector per incident:[reassignment_count, reopen_count, sys_mod_count, priority, impact, urgency]Data is standardized using StandardScaler and projected into 2D using IncrementalPCA.This allows detection of structural clusters and operational anomalies.

Figure 1: PCA Projection (Priority Encoded using Sequential Oranges)

3. Escalation Impact Analysis

Resolution Time (hours) = (closed_at − opened_at)Incidents grouped by reassignment_count (0,1,2,3+).Mean resolution time updated incrementally.Sequential Blues encoding reflects escalation intensity.

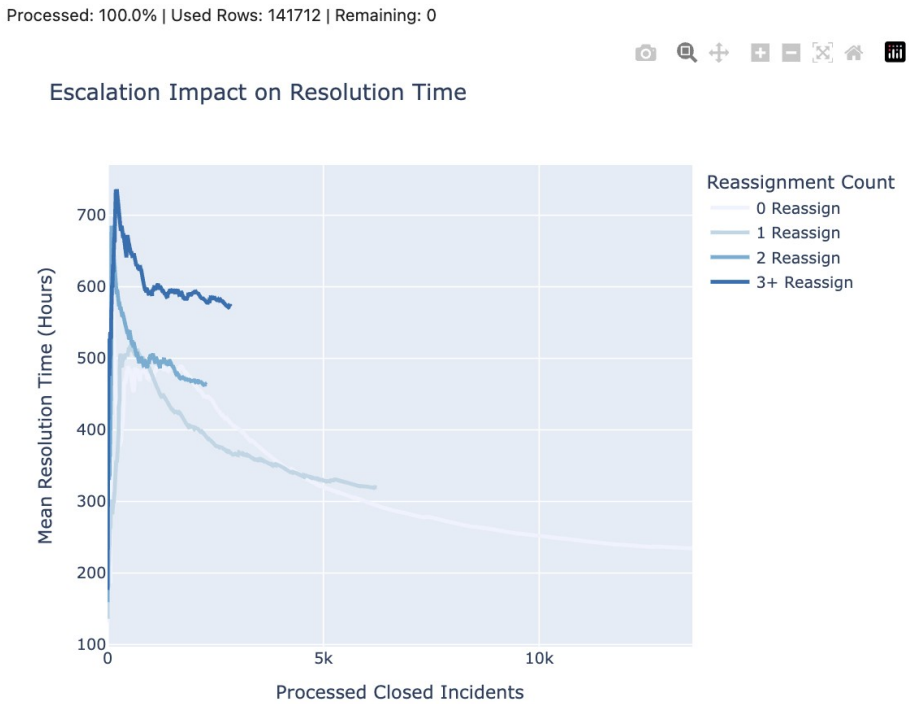

Figure 2: Escalation Impact Dashboard

4. Priority Impact Analysis

Priority values (1–5) extracted numerically.Streaming mean resolution time computed per priority.Sequential Oranges encoding preserves perceptual hierarchy.

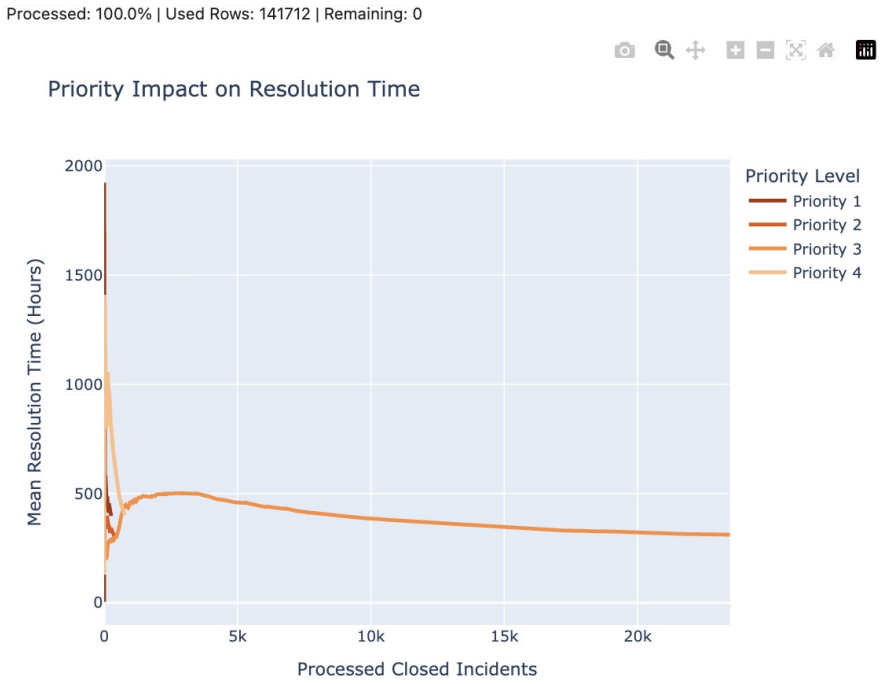

Figure 3: Priority Impact Dashboard

5. SLA Breach Analysis

SLA Breach Rate = (# made_sla == False) / Total × 100Grouping supported by Priority, Escalation, and Category.Color encodings follow ColorBrewer standards (Sequential + Qualitative).

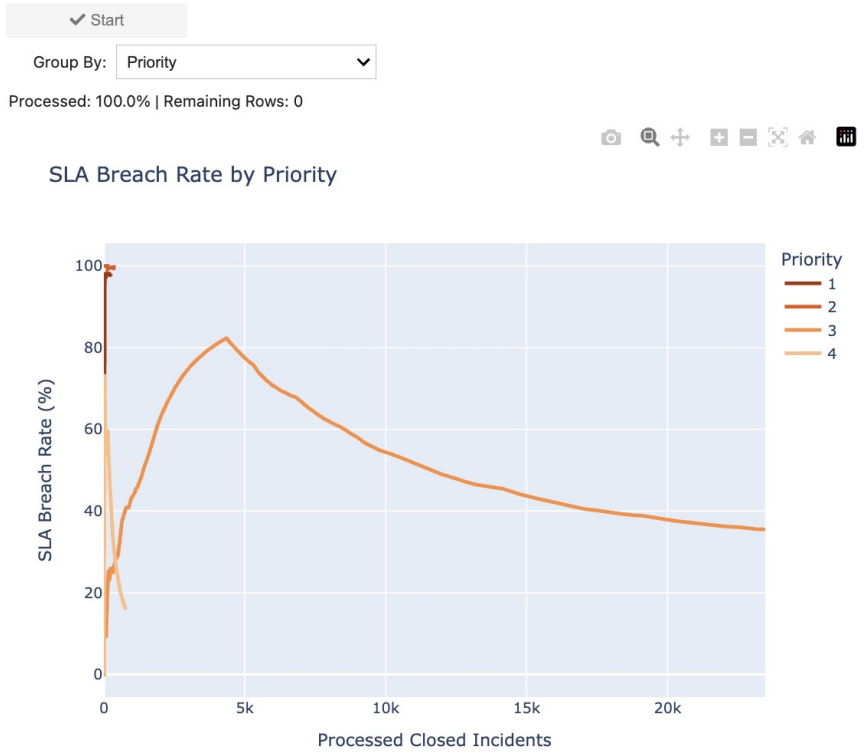

Figure 4: SLA by Priority

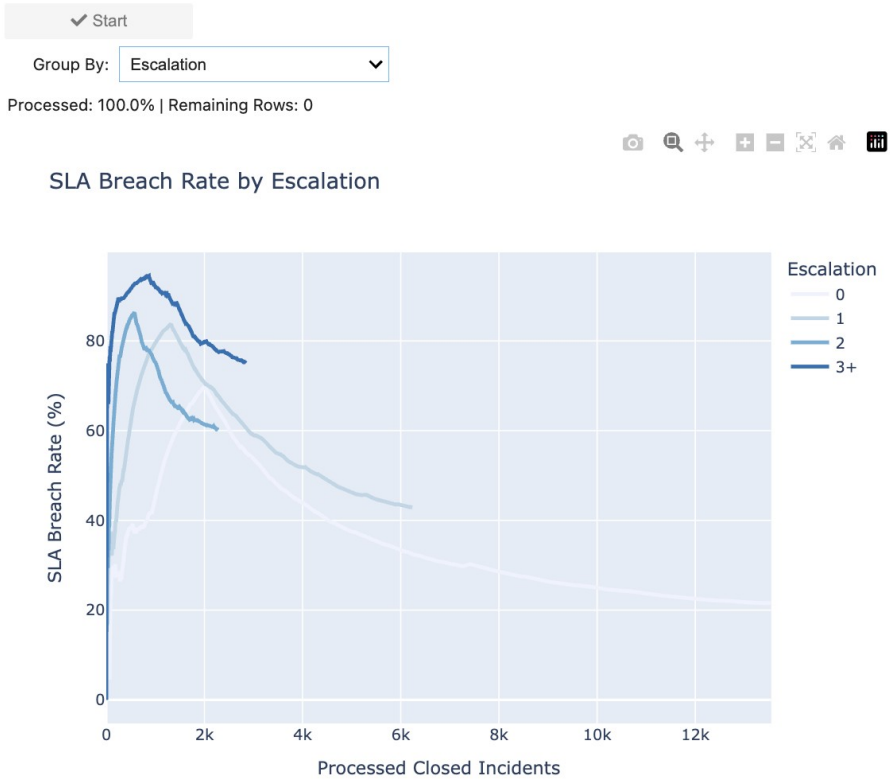

Figure 5: SLA by Escalation

Figure 6: SLA by Category

6. Forensic Case Study – May 2016

Date Range: 1 May 2016 – 31 May 2016Boxplot distribution includes median, IQR, and outliers.Observed anomaly:Priority 1 median = 190.17 hoursPriority 2 median = 141.43 hoursThis contradicts expected SLA urgency ordering and indicates structural bottlenecks.

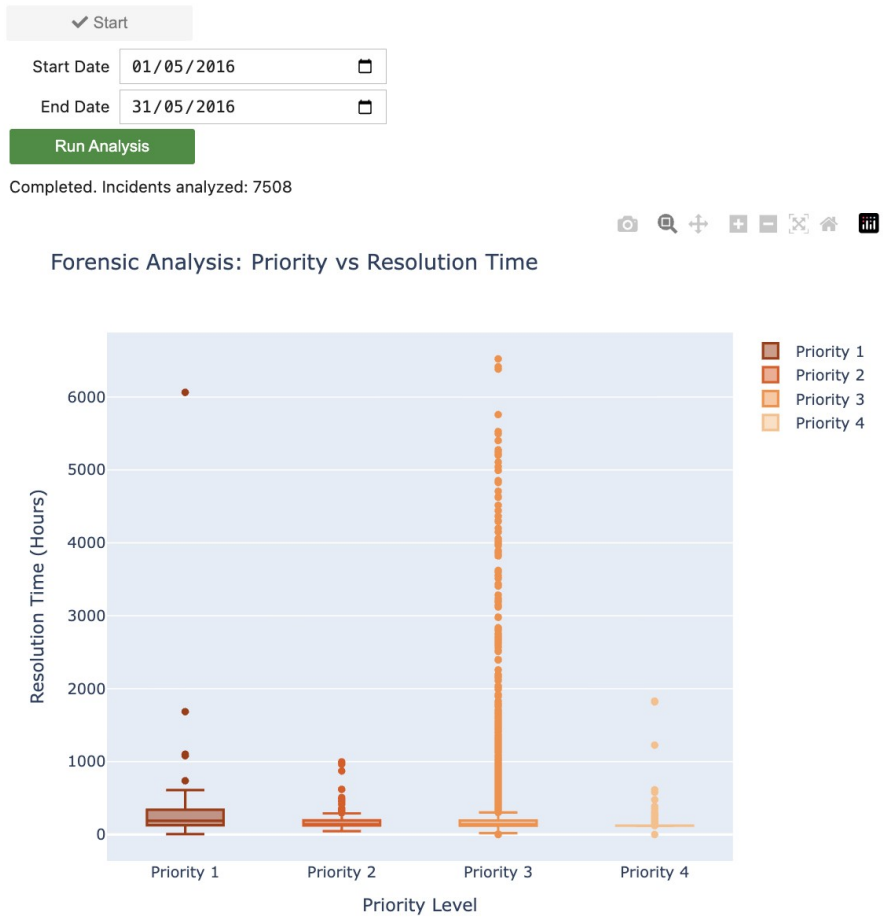

Figure 7: Priority Distribution (May 2016)

7. Statistical Validation

Independent Pandas aggregation confirms visual results.Median resolution times and SLA breach rates were validated directly from filtered dataset.

8. Evaluation Compliance

✔ Dimensionality reduction integrated ✔ Coordinated visualizations implemented ✔ Interactive analytics supported ✔ Standard color encodings applied ✔ Legends included ✔ Documented insights and anomaly explanation

Conclusion

The system demonstrates a complete progressive visual analytics pipeline combining dimensionality reduction, interactive dashboards, statistical validation, and forensic anomaly investigation.